

Summary

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We know, from the Antiderivatives section, that

(1) the position function, $s(t)$, of the object moving along a straight line with velocity $v(t)$ is an antiderivative of the velocity, i.e.

$$\int v(t) dt = s(t) + C$$

(2) the displacement of the object over the time interval $[a, b]$ is given by

$$\text{Displacement} = \Delta s = s(b) - s(a)$$

In this section, we learned that

(3) the displacement of the object over the time interval $[a, b]$ can be represented as the area under the curve $y = v(t)$, (if $v(t) \geq 0$ on $[a, b]$), which means that

$$\text{Displacement} = \int_a^b v(t) dt$$

(4) Combining the last two equations, we get

$$s(b) - s(a) = \int_a^b v(t) dt$$

Note: The last formula states that a definite integral of a *velocity* function can be computed by using an antiderivative of the function and evaluating it at the end points of the interval. This is the first time that we make a connection between definite integrals and antiderivatives.